



HEALTHMATE

AI-Assisted Personalized Health Monitoring and Wellness Management System

Project Proposal

Your family's health, in one place.

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PART A

THE PROBLEM

1. Project Overview

HealthMate is a personalized health companion rather than an AI doctor. It organizes relevant health and wellness information, analyzes personal trends, provides context-aware guidance, identifies potentially unusual patterns, and connects users with trusted family or healthcare support when appropriate.

The product philosophy is simple: health information should not remain passive data. Where appropriate, it should be converted into understandable information, timely reminders and responsible next-step guidance.

Vision. To become the daily health companion for Bangladeshi families, helping people understand their own health and stay connected to the health of the people they love.

Terminology. This proposal describes HealthMate as a *health and wellness companion*. The term "medical assistant" is deliberately avoided because it implies clinical decision-making, which is outside the product's boundary (Section 12).

2. Background and Real-World Context

This section sets out the real-world context, and the market it sits in, before Section 3 states the specific problems that follow from it.

2.1 Background

Users increasingly obtain health-related information from several sources: manual records, smartphones, fitness applications, wearable devices and home-monitoring equipment. Access to data does not mean that users understand what individual measurements mean in relation to their own long-term patterns. A person may know today's blood pressure yet fail to notice a gradual rise over several weeks.

Health information also becomes fragmented across applications, devices and paper records, and family context is rarely connected to personal monitoring. The challenge is greater when elderly family members need additional attention while their children or caregivers live separately.

HealthMate addresses this context by bringing personal monitoring, family health history, lifestyle tracking, longitudinal analysis, personalized guidance, elderly monitoring, healthcare navigation and emergency-aware support into one platform.

2.2 Market Context

The figures below come from public sources listed in the References section and were accessed on September 20, 2026. Several are provisional or are news reports of official data, so the original publications should be consulted for final citation.

Indicator	Value	Source
Population aged 60 and above	About 15.3 million people, around 9.28% of the population (Census 2022), up from 7.48% in 2011. UNFPA projects about 36 million by 2050	[1]
Internet users	77.7 million users, 44.5% of the population (January 2025)	[2]
Bangladeshi workers going abroad	1,130,757 workers migrated for employment in 2025 (up from 1,011,969 in 2024). BMET data reported by the press indicates more than 13 million have gone abroad since 1976	[3], [4]
Remittance inflow	US\$35.56 billion in FY2025-26 (provisional), up 17.3% from US\$30.32 billion the previous year	[5]
Hypertension among adults (18+)	27.4% crude prevalence (BDHS 2017-18). Only 42.5% of those affected were aware of it, and 37.0% were on treatment	[6]
Diabetes among adults (20-79)	13.9 million adults, 13.2% age-standardised prevalence (2024), of whom 39.1% are undiagnosed. Projected 23.1 million by 2050	[7]
Local digital health apps	Telemedicine apps already have sizable user bases (for example, Sebaghar reported over 500,000 downloads in 2024) but focus on consultations, appointments and medicine delivery	[8]

These figures point to three conclusions. First, a large and growing older population, with high rates of hypertension and diabetes that are often undiagnosed or poorly controlled, creates a real need for routine monitoring and early awareness. Second, a very large number of Bangladeshis work abroad and send money home, and many have parents living in Bangladesh, which is the situation the family-alert feature is designed for. Third, existing local apps concentrate on booking and consultation rather than continuous monitoring with family-aware alerts.

Market sizing approach. The total addressable market (TAM) is internet-using adults in Bangladesh (77.7 million users [2]). The serviceable market (SAM) is smartphone users who are, or have parents who are, aged 40 and above. The obtainable market (SOM) is the first-year adoption within the launch segment (Section 5.3), to be set after user interviews and a pilot.

3. Problem Statement

Building on this context, the problems below are the specific pain points HealthMate is designed to solve. Each is assigned an ID so that it can be traced to objectives, requirements, design components and test cases in later parts and in Appendix B.

ID	Problem	HealthMate response
P-01	Fragmented health information. Users obtain information from different sources, making it hard to keep a unified view of their health.	Centralized personal health profile and longitudinal health timeline.
P-02	Limited personalization. Many systems display measurements without considering age, lifestyle, goals, previous patterns or family context.	Context-aware personalized wellness guidance.
P-03	Difficulty recognizing gradual changes. A single measurement may not reveal a meaningful long-term change.	Historical analysis and comparison with the user's established pattern.
P-04	Disconnected family health context. Relevant family history may be forgotten or unused in personal monitoring.	Family history used as a contextual factor for health awareness.

ID	Problem	HealthMate response
P-05	Difficulty maintaining healthy habits. General advice is easy to ignore when it is not connected to actual behavior.	Context-aware reminders for hydration, sleep, activity and wellness goals.
P-06	Elderly monitoring gap. Families have difficulty monitoring older relatives who live separately.	Consent-based caregiver alerts for selected events.
P-07	Uncertainty about professional care. Users may not know when to seek advice or which specialist category is appropriate.	General healthcare-navigation guidance without diagnosis.
P-08	Emergency communication gap. A sudden incident is harder to manage when the affected person cannot respond, for example while travelling.	User verification, emergency-contact notification and authorized location sharing where supported.
P-09	Poor preparation for consultations. Patients forget recent measurements and symptoms.	Health timeline and doctor-visit summary.

4. Existing Solution Gap

Before presenting HealthMate's own solution in Part B, this section checks whether existing tools already solve the nine problems above.

Digital-health technologies make personal health tracking and wearable monitoring feasible, but users may still face fragmented information, limited personalization, difficulty translating measurements into action, privacy concerns and limited family-centered support. HealthMate focuses on the gap between health-data collection and meaningful health action.

4.1 Common Approaches and Their Limitations

Existing approach	Common limitation	HealthMate direction
Fitness and health trackers	Strong tracking, but broader context may be limited	Connect measurements with personal, family and lifestyle context
Separate health apps	Information can remain fragmented	Centralized personal health profile and longitudinal view
Generic health advice	May not reflect individual patterns or goals	Context-aware personalized wellness notifications
Manual family checking	Often reactive and dependent on direct communication	Consent-based caregiver alerts
General AI assistants	May lack structured longitudinal user context	Controlled AI interaction grounded in available HealthMate data

4.2 Competitor Capability Comparison

Capability	Apple Health / Google Health	Fitness apps (e.g., Samsung Health)	General AI chatbots	Local health apps (e.g., Sebaghar, LifePlus, Zaynax)	HealthMate
Personal measurement tracking	Yes	Yes	No	Partial	Yes
Long-term trend analysis	Partial	Partial	No	Limited	Yes
Uses family health history	No	No	Only if re-entered each time	Limited	Yes
Consent-based family alerts	Partial (tied to one device ecosystem)	Limited	No	None identified	Yes (core feature)
Elderly-friendly simplified experience	Not a focus	Not a focus	No	Limited	Yes
Bangla-language support	Limited	Limited	Partial	Varies	Yes
Conversational assistant using the user's own records	No	No	Not with persistent records	Limited	Yes
Emergency contact escalation workflow	Limited (device-dependent)	Limited	No	Limited	Yes (basic in MVP)
Doctor consultation and e-prescription	No	No	No	Yes	No (out of scope)

Google announced in May 2026 that Google Fit is being retired and replaced by the Google Health app, which also replaces the Fitbit app [9]. Feature sets change quickly, so this comparison reflects the sources reviewed on September 20, 2026.

No existing option in this comparison combines personalized, longitudinal, family-aware and safety-bounded guidance in one Bangla-friendly product. This is the gap Part B addresses.

PART B

THE SOLUTION

5. Product Positioning and Differentiation

The gap identified in Section 4 sets the bar HealthMate must clear. Before the detailed solution is presented in Section 6, this section states in one place what makes HealthMate different and how it will reach its first users.

5.1 Why HealthMate Is Different

HealthMate's advantage is not a single feature. It is the combination of personal context, family awareness and safe guidance in one system that the existing options do not provide together.

Dimension	Typical alternative	HealthMate
Personalized rather than generic	The same advice for every user	Guidance based on profile, goals, lifestyle and family context
Longitudinal rather than single-reading	Shows the latest measurement	Compares readings with the user's own baseline and long-term trend
Family-aware	Data stays with one user	Consent-based family and caregiver alerts, and family-assisted entry
Supportive of older adults	Complex interface, English only	Simple Bangla and English interface, accessibility mode, caregiver support
Alert, guidance and navigation in one system	Separate tracker, chatbot and booking apps	One flow from measurement to alert to next-step guidance
Safety-bounded	Unrestricted advice	Explainable rule-based alerts, constrained AI, no diagnosis

HealthMate supports older adults as one important design principle, but it is built for the whole family: adults tracking their own health, older parents, and the relatives who care for them.

5.2 Positioning Statement

For individuals and families who want to understand and support their health, HealthMate is a health and wellness companion that turns everyday measurements into personalized guidance and family-aware alerts. Unlike general fitness trackers and general-purpose AI chatbots, HealthMate connects personal history, family context and consent-based family support in one Bangla-friendly platform.

5.3 Market Entry Strategy

A broad vision needs a focused entry point. The proposed sequence is:

Stage	Segment	Rationale
Stage 1 (capstone MVP and pilot)	Adults with parents or relatives who require health monitoring, including families separated by distance	Clear pain, strong motivation to adopt, clear value from family alerts
Stage 2	Individuals with hypertension, diabetes or other conditions requiring routine monitoring	Regular data entry, high retention potential
Stage 3	General adult population seeking daily wellness support	Broader market once trust and usage are established

6. Proposed Solution

With differentiation and market entry established, this section describes how HealthMate actually works.

HealthMate collects or receives user-provided health and lifestyle information, organizes it into a longitudinal profile, analyzes patterns, and generates appropriate wellness guidance and alerts.

6.1 Core Workflow

```
Personal Information + Family Health History + Health Measurements  
+ Lifestyle Data + Historical Records  
↓  
HealthMate Analysis Engine  
↓  
Trend Analysis + Personalized Wellness + Abnormal-Pattern Awareness  
↓  
User Guidance + Family Alerts + Healthcare Navigation
```

6.2 Conceptual Architecture

The figure shows the conceptual structure of the system. Users provide or share data, an analysis layer interprets it, and an action layer turns the result into alerts, guidance or an escalation workflow, all under cross-cutting security and privacy controls. Part 2 presents the detailed architecture.

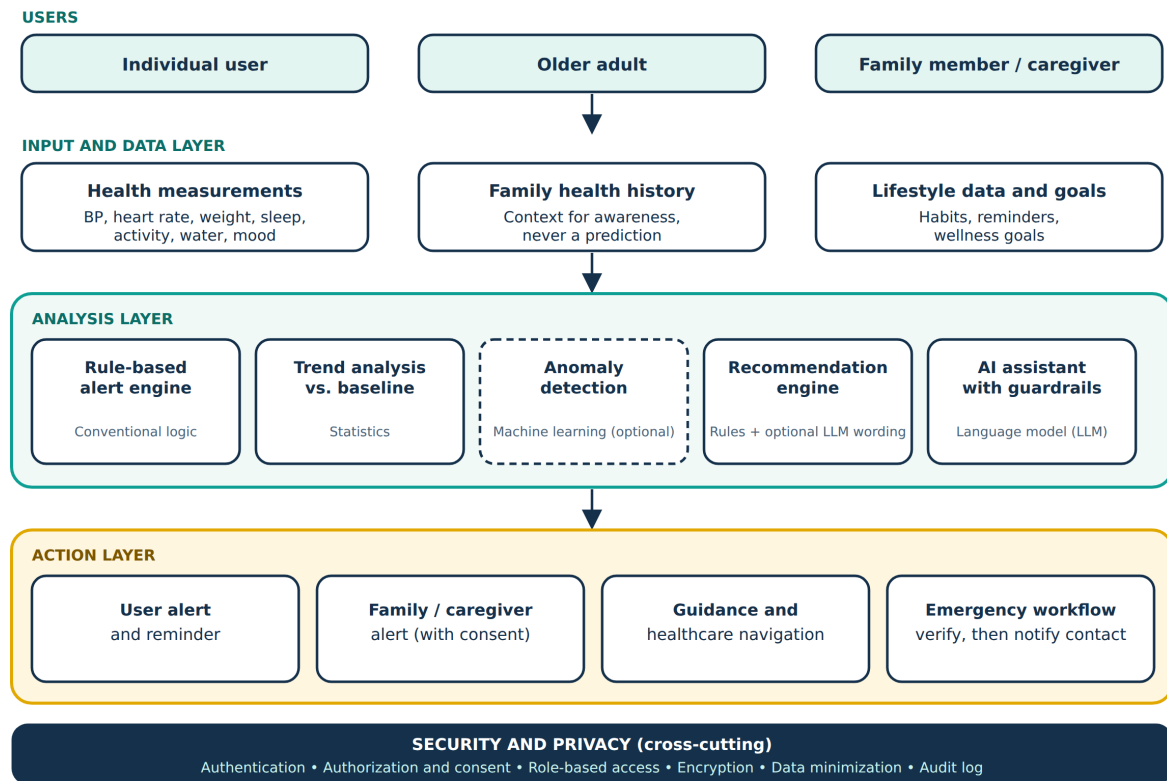


Figure 1. Conceptual architecture of HealthMate. Dashed border marks an optional component.

6.3 Alert Severity Model

HealthMate uses five alert levels. Severity is decided by predefined rules, and the recipients are decided by the user's consent settings.

Level	Name	Typical trigger	System behavior	Recipients
0	Normal	Reading and behavior within the user's expected pattern	No alert. Positive feedback where relevant	User
1	Wellness Reminder	Routine lapse or goal reminder (hydration, sleep, activity, missed reading)	Friendly, personalized nudge	User
2	Attention	Minor deviation from the personal baseline	In-app notification with a non-diagnostic message	User
3	Concern	Significant or persistent deviation, or an out-of-range reading	Notification with a suggestion to consult a professional. Family alert if authorized	User and authorized family or caregiver
4	Emergency escalation	Critical reading or reported symptom, unanswered safety check-in, or manual trigger	Verification prompt, then emergency-contact notification, with location if authorized	User, then emergency contacts

Emergency principle. HealthMate does not clinically determine an emergency. It detects predefined abnormal signals or events and initiates a configured escalation workflow.

6.4 Data Entry Strategy

Manual data entry is a known adoption risk, particularly for older adults. The product addresses it through:

- quick-entry interfaces requiring the fewest possible taps;
- scheduled reminders at the user's preferred times;
- **family-assisted entry**, where an authorized caregiver records readings on behalf of an older adult; and
- optional import from devices and wearables in future versions.

7. Intelligent System Strategy

The architecture in Section 6.2 referred to an analysis layer with several components. This section states exactly which of them use artificial intelligence, and which do not, so that the design remains explainable.

HealthMate does not use AI as a project label. Each component is selected according to the problem it is intended to solve, and explainable methods are preferred. The processing pipeline runs from rule-based logic, to statistical analysis, to anomaly detection, to the recommendation engine, and finally to the language-model assistant.

Component	Category	Approach	Role in HealthMate
Rule-based alert engine (MVP core)	Conventional logic	Configurable thresholds derived from published clinical guidance and reviewed by a qualified healthcare professional before deployment	Classifies readings into alert levels from the first entry. Every alert can state the rule that triggered it
Statistical trend analysis (MVP core)	Statistics	Moving averages, rate of change and deviation from the personal baseline	Reveals gradual changes. Works with a small amount of data
Anomaly detection (enhancement)	Machine learning	A method such as Isolation Forest once enough history exists, validated on test or synthetic data against the rule-based baseline	Flags "unusual compared with your recent pattern". Never a diagnosis
Recommendation engine (MVP)	Conventional logic, with optional generative wording	Rule-based selection from user context, goals and trends, with optional language-model rewording	Personalized reminders that remain controllable
AI health assistant (MVP)	Generative AI (language model)	A language model operating behind a guardrail layer	Explains the user's records and supports healthcare navigation

Only two of the five components use artificial intelligence: anomaly detection (machine learning, optional) and the language-model assistant (with optional wording in the recommendation engine). The alert engine, trend analysis and recommendation selection are conventional software logic and statistics. This separation keeps the system explainable and makes clear where AI is, and is not, used.

The **guardrail layer** consists of input classification (for example emergency, diagnosis request or medication request), constrained prompts, output checks for prohibited content, mandatory disclaimers, and redirection to emergency help when emergency language is detected. Data sent to any external AI service is minimized and free of direct identifiers wherever possible.

PART C

WHO IT IS FOR

8. Target Users and Stakeholders

Having described what HealthMate does, this section defines who it is for.

8.1 Target Users

User type	Description	Key needs
Primary user (self-managing adult)	Adult who wants to track health, habits and goals	Simple recording, clear trends, useful reminders
Older adult	Person aged approximately 60 and above, possibly with chronic conditions	Large text, Bangla interface, minimal steps, ability to receive help from family
Family member / caregiver	Adult child, relative or caregiver, possibly living abroad	Timely, relevant alerts. No overwhelming detail. Clear permission boundaries
Healthcare professional (secondary, future)	Doctor or nurse	Structured health summary
Healthcare ecosystem partners (future)	Hospitals, telemedicine, device providers, emergency services	Standard integration interfaces

Illustrative personas, to be validated through user interviews:

- **Rahim, 32**, works in another country. His father, 64, has hypertension and lives in Dhaka. Rahim wants to know if something changes without calling every day.
- **Mrs. Sultana, 61**, measures her blood pressure occasionally, prefers Bangla and is not comfortable with complex apps.
- **Nabila, 27**, wants to improve sleep, hydration and activity, and her family has a history of hypertension.

8.2 Stakeholders

Stakeholder	Role	Interest	Influence
Individual user	Primary user	Understand and improve own health	High
Older adult	Primary user	Simple, safe monitoring	High
Family member / caregiver	Alert recipient	Timely awareness of a relative's health	High
Project team	Design, development, testing	Deliver a working, well-documented system	High
Project supervisor	Academic guidance and evaluation	Quality, rigor and scope control	High

Stakeholder	Role	Interest	Influence
Healthcare professional (future)	Potential recipient of summaries	Accurate, well-structured information	Medium
Hospital / provider (future)	Potential integration partner	Referral and data exchange	Low (currently)
Emergency services (future)	Potential escalation partner	Reliable, verified information	Low (currently)
Regulators	Oversight of health data and health software	Compliance and safety	Medium
Future investors / business partners	Funding and scale	Market, traction and sustainability	Medium (future)

9. Use-Case Scenarios

The scenarios below show the users of Section 8 and the solution of Section 6 working together in practice. Names are illustrative and will be validated through user interviews.

Scenario	Situation	How HealthMate helps	Key requirements
1. Young user with family history	Nabila, 27, has a family history of hypertension. Her own readings are normal	Records the family history as context, suggests periodic blood-pressure checks, and sends reminders about activity, sleep and hydration. It does not predict disease	FR-10, FR-12, FR-13, FR-40
2. Elderly parent living alone	Mrs. Sultana, 61, lives in Dhaka. Her son Rahim works abroad	She records readings with a quick-entry screen, or a caregiver enters them. When two days of readings rise, or a routine check-in is missed, Rahim receives an alert that shows only what she has authorized	FR-22, FR-23, FR-60 to FR-64, FR-66
3. Critical reading and no response	An older user records a reading in a critical range	The system reaches Level 4, asks the user to confirm their status, and if there is no reply within the set time notifies the emergency contact, with the last known location if the user has authorized it	FR-33, FR-70 to FR-75
4. Gradual long-term change	A user's blood pressure rises slowly over six weeks, and every single reading looks acceptable	Trend analysis against the personal baseline moves the alert from Attention to Concern, explains the change in non-diagnostic language, and suggests seeing a general physician. A visit summary can be prepared	FR-31 to FR-33, FR-52, FR-81

Automatic detection of incidents while a person is travelling or driving is not part of the prototype. It would require wearable or vehicle data and is listed as future scope (Section 11.3).

PART D

GOALS, SCOPE AND REQUIREMENTS

10. Project Objectives

The scenarios in Section 9 describe the experience HealthMate should deliver. This section turns that experience into measurable objectives.

10.1 General Objective

To design and develop an intelligent personalized health monitoring and wellness management system that helps users understand health patterns, receive personalized wellness guidance, identify potentially unusual changes, and connect with trusted family or healthcare support when appropriate.

10.2 Specific Objectives

The five objectives below are measurable. The targets are proposed and will be confirmed with the supervisor together with the testing approach.

ID	Objective	Measure	Target
O-01	Provide fast, easy health recording for general and older users	Acceptance tests of core recording scenarios, and usability sessions with at least 8 participants	100% of scenarios pass. Median time to record a reading under 30 seconds. At least 85% task completion and a SUS score of at least 70
O-02	Detect meaningful trends and unusual readings, assign the correct alert level, and provide personalized reminders	Trend and rule-engine outputs checked against reference test cases (Levels 0 to 4), and reminders checked across defined user profiles	100% of defined test cases match the expected result. Reminders correct across at least 10 profile scenarios
O-03	Provide a safe AI assistant	Safety test set, for example requests for diagnosis or medication changes	100% of prohibited requests receive a safe redirect
O-04	Provide consent-based family alerts and a basic emergency workflow	Access-control tests, and emergency scenario tests (verification, then contact notification)	0 unauthorized disclosures. Escalation completes in all defined scenarios
O-05	Protect sensitive data and stay modular for future integrations	Security review, documented interfaces and a simulated device feed	No critical or high-severity findings open. Each engine replaceable without changing the others. One simulated feed accepted through the interface

11. Scope and Boundaries

Not every capability that supports these objectives can be built in one semester. This section defines what is, and is not, part of the capstone MVP.

11.1 Initial Capstone MVP

- User registration, authentication and profile management
- Personal health profile and family health history
- Health-data recording, dashboard and historical records
- Trend analysis and rule-based alerts with alert levels
- Personalized wellness reminders and recommendations
- AI health assistant with safety guardrails
- Symptom-based healthcare navigation (category of professional only)
- Elderly monitoring, family and caregiver authorization, family alerts and family-assisted entry
- Privacy and permission management, and audit logging
- Basic emergency-aware escalation workflow (user verification and emergency-contact notification)
- Bangla and English interface for core screens
- System testing and evaluation

11.2 Planned Enhancements (Time Permitting)

- Health timeline and doctor-visit summary
- Medication and appointment reminders
- Anomaly detection using a machine-learning model
- More advanced personalization

11.3 Future Scope

- Smartwatch, wearable and IoT device integration
- Continuous real-time monitoring and automatic incident detection
- Advanced time-series prediction and fall-risk monitoring
- Telemedicine and e-prescription
- Hospital, healthcare-provider and emergency-service integration
- Voice-based Bangla interaction
- Large-scale healthcare ecosystem integration

The prototype demonstrates the emergency workflow through software-triggered and manually triggered events. It does not claim automatic detection of incidents and does not connect to real emergency services.

12. Project Boundary, Safety and Regulatory Considerations

The scope above deliberately excludes clinical functions. This section states that boundary explicitly, together with the security, privacy and regulatory controls that make it enforceable.

HealthMate maintains a strict boundary between software assistance and clinical decision-making.

12.1 The System Will and Will Not

- **The system will:** analyze user-provided information, identify potentially unusual patterns, provide wellness guidance, support healthcare navigation, notify authorized family members, and support emergency-aware workflows.
- **The system will not:** diagnose diseases, prescribe medication, instruct users to stop prescribed treatment, replace doctors, make autonomous clinical decisions, or claim certainty from imperfect consumer health measurements.

HealthMate does not clinically determine an emergency. It detects predefined abnormal signals or events and initiates a configured escalation workflow.

12.2 Security and Privacy Framework

Sensitive health information is handled with the following controls, each traceable to requirements.

Control	What it means for HealthMate	Requirements
Authentication	Verified accounts, secure login, password reset and secure sessions	FR-01 to FR-03, NFR-02
Authorization	Server-side enforcement of who may see or change each record	FR-04, NFR-03
Consent	Explicit, revocable, per-category consent for family sharing and emergency notification	FR-60 to FR-63, FR-70
Role-based access	Individual, caregiver and administrator roles with least privilege	FR-04, NFR-03
Encryption	TLS for data in transit and encryption of sensitive data at rest	NFR-01
Data minimization	Only necessary data is collected or sent to external services. Users can export and delete their data	NFR-04
Audit and logging	Access to shared health data is recorded and visible to the data owner	FR-65, NFR-05

These controls follow four principles: consent, purpose limitation, minimization and transparency.

12.3 Regulatory Considerations

The main legal development relevant to HealthMate is Bangladesh's **Personal Data Protection Ordinance, 2025 (Ordinance No. 61 of 2025)**, issued on November 6, 2025 as the country's first standalone personal-data-protection law [10]. According to published summaries, it treats individuals as owners of their personal data, requires explicit consent, applies stricter rules to sensitive personal data such as health records and to cross-border transfers, and creates enforcement through a national data authority, with a compliance period of about 18 months for some provisions.

HealthMate's design is consistent with these principles. Explicit and revocable consent for family sharing (FR-60 to FR-63), data export and deletion (NFR-04), and minimization of data sent to external AI services address the main obligations. The team will confirm the ordinance's current legal status and commencement dates, and whether any medical-device, telemedicine or digital-health guidelines apply to a wellness application of this type, before the final proposal. The prototype uses no real patient data. Testing uses synthetic data or data from consenting volunteers.

13. Requirements Analysis

The objectives, scope and safety boundary above are now translated into concrete, testable requirements.

Requirements are identified by ID and prioritized with four levels: **Must have** (capstone MVP), **Should have**, **Could have** (planned enhancement) and **Future** (outside the capstone, see Section 11.3). The full functional requirement list is in Appendix A.

13.1 Functional Requirements (Summary)

Module	IDs	Main capabilities	Priority
Account and access	FR-01 to FR-04	Registration, secure login, password reset, roles (individual, caregiver, administrator)	Must
Health profile	FR-10 to FR-13	Personal profile, BMI, existing conditions, family health history used only as context	Must
Health data recording	FR-20 to FR-24	Blood pressure, heart rate, weight, sleep, activity, water, mood. Validation, quick entry, family-assisted entry	Must
Dashboard, trends and alerts	FR-30 to FR-35	Dashboard, trend periods, personal baseline, rule-based alert levels, non-diagnostic wording	Must
Personalized guidance	FR-40 to FR-42	Personalized reminders, wellness goals, configurable reminder times	Must / Should
AI assistant and navigation	FR-50 to FR-55	Natural-language questions, explanation of records, specialist-category guidance, refusal of diagnosis requests, safety checks	Must
Family monitoring and permissions	FR-60 to FR-66	Invite caregivers, per-category permissions, revocation, family alerts, audit log, missed check-in alert	Must / Should
Emergency-aware escalation	FR-70 to FR-75	Emergency contacts, user verification prompt, contact notification, optional location, manual trigger, escalation log	Must / Should
Consultation support	FR-80 to FR-82	Health timeline, doctor-visit summary, medication and appointment reminders	Could
Localization	FR-90 to FR-91	Bangla and English interface, accessibility mode for older adults	Must / Should

13.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	All data in transit shall be encrypted (TLS). Sensitive health data at rest shall be encrypted.
NFR-02	Security	Passwords shall be stored using a modern salted hash. Authentication shall use secure session or token management.
NFR-03	Access control	Access to any health record shall be enforced server-side through role-based access and explicit user consent.
NFR-04	Privacy	The system shall collect only the data required for its functions (data minimization). Users shall be able to export and delete their data.
NFR-05	Auditability	Access to health data by anyone other than the owner shall be logged.
NFR-06	Performance	Dashboard pages shall load within 3 seconds under normal load. Alert evaluation shall occur within 10 seconds of data submission.

ID	Category	Requirement
NFR-07	Reliability	Alert delivery shall be retried on failure. Failed deliveries shall be recorded.
NFR-08	Usability	Recording a routine measurement shall be completable in a small number of steps and within the target in O-01.
NFR-09	Accessibility	The interface shall follow accessibility good practices for older users (font size, contrast, touch target size).
NFR-10	Localization	All user-facing text shall be externalized to support multiple languages.
NFR-11	Maintainability	The system shall be modular, with documented interfaces between components, and automated tests for core logic.
NFR-12	Scalability	The architecture shall allow horizontal scaling of stateless services and shall not depend on a single-user-per-server design.
NFR-13	Safety	Alert and assistant behavior shall comply with the boundary defined in Section 12.
NFR-14	Cost awareness	Cost per active user (hosting, notifications, AI API usage) shall be estimated in the design phase.

13.3 Feature-to-Problem Mapping

The table below closes the loop back to Section 3: for each HealthMate feature, the real-life problem it addresses and the value it provides.

HealthMate feature	Real-life problem	Proposed value	Problem ID
Personal health profile	Health information is scattered	Centralized personal health context	P-01
Family health history	Family health risks may be ignored	Better health awareness	P-04
Health monitoring	Users lack organized health records	Continuous personal tracking	P-01
Trend analysis	Gradual changes are difficult to notice	Better understanding of long-term patterns	P-03
Abnormal-pattern detection	Unusual changes may go unnoticed	Early awareness	P-03
Personalized notifications	Generic advice is easy to ignore	Context-aware guidance	P-02, P-05
AI health assistant	Users may struggle to understand health information	Easier interaction	P-07
Healthcare navigation	Users may not know which specialist to consult	Better healthcare direction	P-07
Elderly monitoring	Children may live far from elderly parents	Remote family awareness	P-06
Family alerts	Important changes may remain unnoticed	Faster family response	P-06
Permission system	Family monitoring may create privacy concerns	User-controlled sharing	P-06
Health timeline	Health history may be scattered	Organized longitudinal record	P-01, P-09
Doctor-visit summary	Users may forget previous measurements	Better preparation for consultation	P-09
Medication and appointment support	Users may forget routine healthcare tasks	Better adherence support	P-05, P-09

HealthMate feature	Real-life problem	Proposed value	Problem ID
Emergency escalation	Unresponsive users may be difficult to reach	Faster communication pathway	P-08

PART E

HOW WE WILL BUILD IT

14. Feasibility Analysis

With the requirements defined, this section checks whether they can realistically be delivered.

14.1 Feasibility Assessment

Dimension	Assessment
Technical	Feasible using standard web and mobile technologies, a relational database, rule-based logic, statistical methods and an external language-model API. The design avoids complex clinical AI that requires large proprietary datasets.
Schedule	Feasible with iterative delivery. Core monitoring and alerts are prioritized first. Enhancements are deferred if needed so that a working MVP is always available.
Economic	Feasible using open-source tools, student cloud credits and limited API usage. Hardware and provider integrations are excluded from the MVP.
Operational	Feasible through a simple interface, Bangla support, understandable notifications and user-controlled sharing.
Legal and ethical	Feasible with the boundary in Section 12, use of synthetic or consented data, and no clinical claims. Requires confirmation of applicable regulations.

14.2 What Is Implemented Now and What Remains Future

Feasibility is shown by separating available technology, what the capstone implements, and what stays in future scope. For example, direct wearable integration is future scope, but manual and family-assisted entry and a simulated device feed are part of the MVP.

Capability	Available technology	In the capstone MVP	Remains future scope
Health data input	Entry forms, caregiver entry, integration interface	Manual and family-assisted entry, plus a simulated device feed through the documented interface	Direct integration with Health Connect, smartwatches and Bluetooth blood-pressure devices
Trend analysis	Standard statistical libraries	Moving averages and baseline comparison	Advanced time-series forecasting
Alerts	Configurable rule engine	Threshold rules and five alert levels	Thresholds personalized from real usage data
Anomaly detection	Isolation Forest and similar methods	Evaluated on test or synthetic data (enhancement)	Models trained on real usage data
AI assistant	Language-model API with guardrails	Explanation, navigation and safe refusal	Domain-specific models and Bangla voice interaction
Notifications	Push notifications, SMS gateway	Push notifications and limited SMS for the demonstration	Reliable SMS and voice-call escalation

Capability	Available technology	In the capstone MVP	Remains future scope
Location sharing	Device location with user permission	Optional last known location, permission-based	Continuous tracking
Emergency workflow	Software workflow	Verification prompt and contact notification	Automatic incident detection and emergency-service integration
Healthcare connection	Standard export formats	Doctor-visit summary export (enhancement)	Provider dashboards, telemedicine and hospital integration

15. Development Methodology

Given that the project is feasible, this section explains how the team will deliver it.

Selected SDLC model: Agile (iterative and incremental). Agile is selected because HealthMate contains several interconnected modules and technically uncertain components that benefit from iterative refinement, supervisor feedback, usability testing and incremental validation. It allows the team to keep a working product while progressively adding monitoring, analytics, AI assistance, elderly monitoring and integration capabilities.

Working practices: iterations with a defined goal and deliverable, a prioritized product backlog based on Appendix A, an iteration review with the supervisor, version control with reviewed changes, and automated tests for core logic (alert rules, trend calculations and access control).

Iteration	Focus	Main requirements	Deliverable
1	Foundation	FR-01 to FR-04, FR-10 to FR-13, NFR-01 to NFR-03	Architecture, database schema, authentication, profile
2	Monitoring	FR-20 to FR-24, FR-30, FR-41	Data recording, dashboard, history
3	Analysis and alerts	FR-31 to FR-34, FR-40	Trend analysis, rule-based alert engine, reminders
4	Assistant	FR-50 to FR-55	AI assistant with guardrails, navigation guidance
5	Family and emergency	FR-60 to FR-66, FR-70 to FR-75	Consent management, family alerts, emergency workflow
6	Integration and evaluation	FR-90, FR-91, all NFRs	Localization, security review, usability testing, final evaluation, documentation

16. Resource, Time and Budget Planning

The methodology above is now scheduled against the available 14-week semester and the team's resources.

16.1 Resources

Resource area	Purpose	Planning approach
Development tools	System design and implementation	Prefer open-source and existing academic tools
Database	Health and user data storage	Relational database such as MySQL or PostgreSQL
AI and API services	Conversational and intelligent functions	Limited, controlled usage within budget
Notification services	Push notifications and limited SMS for the demonstration	Free or low-cost tiers
Hosting	Prototype deployment and testing	Basic cloud or server resources, student credits where available
Testing devices	Functional and usability testing	Available team devices. External hardware optional
Development time	Iterative implementation	Agile iterations with MVP-first prioritization

16.2 Time Plan

The project runs over a 14-week semester. Proposal documentation takes place in weeks 1 to 4, and each of the six development iterations is planned at two weeks, starting in week 2 as the design is completed. Usability sessions and evaluation run in weeks 11 to 13, the final report and presentation in weeks 13 to 14, and week 14 is kept as a buffer. A supervisor review is held at the end of each iteration. Because the plan is MVP-first, later enhancements can be deferred without losing a working system.

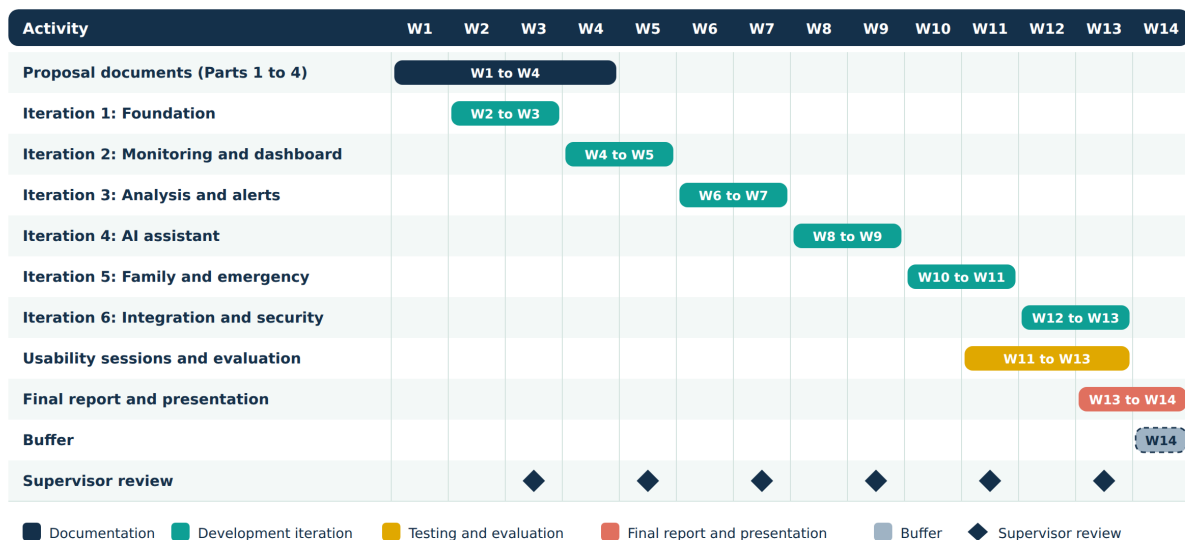


Figure 2. 14-week project schedule (Gantt chart). W = week of the semester.

16.3 Budget Approach

The prototype is designed to run on open-source software, student environments and free or low-cost cloud tiers. The main variable costs are hosting, AI API usage and SMS delivery. AI usage and message volumes are limited to testing and demonstration. A cost-per-active-user estimate will be prepared in the design phase (NFR-14). Costly hardware and external integrations remain future considerations.

17. Risk Analysis

No plan is risk-free. This section identifies what could go wrong in the plan above, and how the team will respond.

Ratings: L = Low, M = Medium, H = High.

ID	Risk	Likelihood	Impact	Mitigation
R-01	Users do not enter data consistently (low engagement)	H	H	Quick entry, reminders, family-assisted entry, usability testing, optional device import later
R-02	Insufficient data for machine-learning anomaly detection	H	M	Rule-based alerts as the core. ML treated as an optional enhancement, tested on synthetic data
R-03	AI assistant gives unsafe or incorrect health information	M	H	Guardrail layer, constrained prompts, prohibited-request test set, disclaimers, emergency redirection
R-04	False alerts cause anxiety or alert fatigue	M	M	Configurable thresholds, alert levels, review of thresholds by a qualified professional, feedback loop
R-05	Missed alerts (a real concern is not flagged)	M	H	Conservative rules, retry logic, explicit statement that the system is not a substitute for medical care
R-06	Privacy breach or unauthorized access to health data	L	H	Encryption, server-side access control, audit logs, security testing, minimal data collection
R-07	Regulatory or legal issues around health software and personal data	M	H	Non-diagnostic boundary, no clinical claims, alignment with the Personal Data Protection Ordinance 2025, review of applicable regulations, synthetic data in testing
R-08	Scope creep due to the breadth of features	H	M	MoSCoW priorities, iteration plan, supervisor scope reviews
R-09	Third-party service limits or outages (SMS, AI API)	M	M	Abstraction layers, fallback providers or mock services, retry logic
R-10	Difficulty recruiting participants for user validation	M	M	Start recruitment early, use family and community networks, define a minimum viable sample
R-11	Team capacity or schedule slippage	M	M	Iterative delivery, defined MVP, regular progress tracking
R-12	Older adults find the interface difficult	M	H	Accessibility mode, Bangla interface, simplified flows, testing with older users

Several of these risks are already reduced by decisions made earlier in this proposal, rather than left for later: **R-01** (inconsistent data entry) is addressed by the quick-entry and family-assisted entry design in Section 6.4; **R-02** (insufficient data for machine learning) is addressed by relying on rule-based alerts as the MVP core rather than machine learning (Section 7); **R-03** (unsafe AI output) is addressed by the guardrail layer (Section 7); **R-06** (privacy breach) is addressed by the security and privacy framework (Section 12.2); and **R-07** (regulatory issues) is addressed by the alignment with the Personal Data Protection Ordinance in Section 12.3.

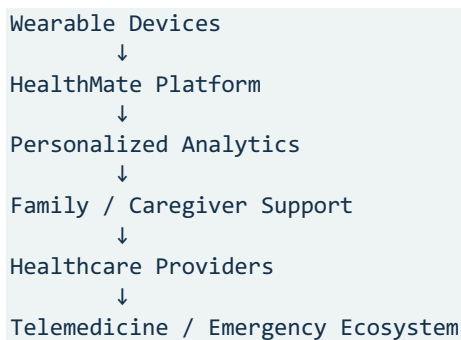
PART F

LOOKING AHEAD

18. Scalability and Product Vision

Beyond the risks of delivering the MVP, this section looks past the capstone toward how HealthMate could grow.

Although the initial HealthMate system is a capstone prototype, its architecture is designed with future scalability in mind.



18.1 Roadmap

Stage	Focus	Example capabilities
Stage 1: Capstone MVP	Core monitoring, alerts, family sharing, assistant	Everything in Section 11.1
Stage 2: Connected data	Reduce manual entry	Blood pressure device and smartwatch integration, import from Health Connect and similar platforms
Stage 3: Deeper intelligence	Improve personalization	Anomaly detection trained on real usage data, advanced time-series prediction, fall-risk awareness
Stage 4: Care ecosystem	Connect to healthcare	Provider-facing summaries, telemedicine, hospital and pharmacy integration
Stage 5: Emergency and scale	Reliable escalation	Emergency-service integration, multilingual and voice-based Bangla interaction, privacy-preserving analytics at scale

18.2 Architectural Provisions for Scale

- Stateless backend services that can be scaled horizontally (NFR-12).
- A documented integration interface so that a new data source, such as a wearable, can be added without changing the core engines (O-05).
- Configurable alert rules and externalized text to support new clinical guidance and languages (NFR-10).

These are future product possibilities rather than promises of the capstone prototype.

18.3 Business Model Outline (Vision)

This outline describes the long-term vision. It is not implemented in the capstone, and all revenue assumptions are hypotheses to be validated through user research.

Element	Hypothesis
Customer	Individuals and families. The family member who wants to support a relative is a likely paying customer
Value	Family-aware health awareness, guidance and alerts in Bangla
Model options	Freemium with a paid family plan (more family members, richer alerts, history export). Possible future B2B partnerships with clinics, telemedicine services and device providers
Costs	Hosting, notification and SMS costs, AI API usage, support, compliance
Key metrics (future)	Weekly active users, retention, families connected per account, alert-to-action rate, cost per active user
Go-to-market (future)	Start with families separated by distance, then people managing chronic conditions, then the general adult population

19. Expected Value and Impact

If the vision above is realized, this is the value it creates for the people HealthMate serves.

19.1 Individual User Value

- Better understanding of personal health trends.
- More consistent wellness habits through timely reminders.
- Greater awareness of potentially unusual changes.
- More organized health information.
- Better preparation for professional consultations.

19.2 Family Value

- Better awareness of elderly family members.
- Consent-based health alerts controlled by the person being monitored.
- Reduced uncertainty when family members live apart.
- Structured communication during potentially concerning situations.

19.3 Healthcare Interaction Value

HealthMate can help users organize relevant information before a professional consultation and provides a foundation for future healthcare-provider integration.

19.4 Product Value

The long-term opportunity lies in connecting monitoring, personalization, prevention awareness, family safety and healthcare navigation in one digital-health platform.

20. Part 1 Conclusion

This planning phase establishes HealthMate as a feasible, user-centered and scalable digital-health project rather than a simple health-data tracking application. The project is grounded in real-life problems involving fragmented health information, limited personalization, difficulty recognizing changes, elderly safety, family communication and healthcare navigation.

The proposed solution combines monitoring, analysis, personalized wellness guidance, responsible AI assistance, family support and emergency-aware workflows, while maintaining clear medical and ethical boundaries.

This planning foundation, together with the requirement identifiers (P, O, FR, NFR and R) and the traceability set out in Appendix B, guides the subsequent parts.

Part	Focus
Part 2	System design: architecture, data model, interaction flows and interface design
Part 3	To be confirmed with the supervisor
Part 4	Consolidated final project proposal, including the executive summary

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Appendix A: Detailed Functional Requirements

Priority key: **M** = Must have (MVP), **S** = Should have, **C** = Could have (planned enhancement). Items marked Future are listed in Section 11.3 and are not in this table. The problem link column refers to the problem IDs in Section 3.

A.1 Account and Access Management

ID	Requirement	Priority	Problem link
FR-01	The system shall allow users to register and verify an account.	M	P-01
FR-02	The system shall allow users to log in and log out securely.	M	P-01
FR-03	The system shall support password reset.	M	P-01
FR-04	The system shall support role types: individual user, family member/caregiver, and administrator.	M	P-06

A.2 Health Profile

ID	Requirement	Priority	Problem link
FR-10	The system shall store age, gender, height, weight, calculated BMI, lifestyle information and wellness goals.	M	P-01, P-02
FR-11	The system shall allow users to record existing health conditions and current medications as user-provided information.	S	P-02
FR-12	The system shall allow users to record relevant family health history.	M	P-04
FR-13	The system shall use family health history only as a contextual input for awareness and personalization, and shall not present it as a prediction of disease.	M	P-04

A.3 Health Data Recording

ID	Requirement	Priority	Problem link
FR-20	The system shall allow recording of blood pressure, heart rate, weight, sleep duration, physical activity, water intake and mood/stress.	M	P-01, P-05
FR-21	The system shall validate entries (valid ranges, units, timestamps) and warn on implausible values.	M	P-01
FR-22	The system shall allow an authorized caregiver to record measurements on behalf of another user.	S	P-06
FR-23	The system shall provide a quick-entry interface designed for minimal steps.	M	P-05, P-06
FR-24	The system shall allow users to edit or delete their own records.	M	P-01

A.4 Dashboard, Trends and Alerts

ID	Requirement	Priority	Problem link
FR-30	The system shall display current measurements, history, trends, goals and notifications on a dashboard.	M	P-01, P-03
FR-31	The system shall compute trends over selectable periods (for example 7, 30 and 90 days).	M	P-03
FR-32	The system shall compare recent readings with the user's own baseline.	M	P-03
FR-33	The system shall classify readings and patterns into five alert levels (Level 0 Normal to Level 4 Emergency escalation) using configurable, rule-based thresholds.	M	P-03
FR-34	The system shall use non-diagnostic language in all alerts.	M	P-03
FR-35	The system shall support anomaly detection using a statistical or machine-learning method once sufficient data exists for a user.	C	P-03

A.5 Personalized Guidance

ID	Requirement	Priority	Problem link
FR-40	The system shall generate personalized reminders (hydration, sleep, activity and goals) based on profile and recorded behavior.	M	P-05
FR-41	The system shall allow users to set and track wellness goals.	S	P-05
FR-42	The system shall allow users to configure reminder times and types.	S	P-05

A.6 AI Health Assistant and Navigation

ID	Requirement	Priority	Problem link
FR-50	The system shall allow users to ask questions about their own records in natural language (Bangla and English).	M	P-01, P-07
FR-51	The assistant shall provide general wellness information and explain the user's recorded data.	M	P-07
FR-52	The assistant shall suggest a category of healthcare professional based on user-described symptoms, and shall not state or imply a diagnosis.	M	P-07
FR-53	The assistant shall decline requests for diagnosis, medication changes or treatment discontinuation, and shall redirect the user to a qualified professional.	M	P-07
FR-54	The system shall apply input and output safety checks to assistant interactions.	M	P-07
FR-55	The system shall show a clear notice that assistant responses do not replace professional advice.	M	P-07

A.7 Family Monitoring and Permissions

ID	Requirement	Priority	Problem link
FR-60	The system shall allow a user to invite and authorize a family member or caregiver.	M	P-06
FR-61	The system shall allow the user to control which data categories each authorized person may view.	M	P-06

ID	Requirement	Priority	Problem link
FR-62	The system shall allow the user to choose whether each authorized person receives alerts only, summaries, or detailed data.	M	P-06
FR-63	The system shall allow the user to revoke access at any time, with immediate effect.	M	P-06
FR-64	The system shall notify authorized family members of Concern and Urgent alerts according to the user's permissions.	M	P-06
FR-65	The system shall record all access to shared health data in an audit log visible to the data owner.	S	P-06
FR-66	The system shall raise an alert for a missed routine check-in, based on a user-configured schedule.	S	P-06, P-08

A.8 Emergency-Aware Escalation

ID	Requirement	Priority	Problem link
FR-70	The system shall allow a user to configure one or more emergency contacts and give explicit consent for emergency notifications.	M	P-08
FR-71	On a Level 4 (Emergency escalation) event, the system shall prompt the user to confirm their status within a configurable time window.	M	P-08
FR-72	If the user does not respond, the system shall notify the emergency contact(s) with the reason for the alert.	M	P-08
FR-73	If the user has authorized it, the system shall include the last known location in the notification.	S	P-08
FR-74	The user shall be able to trigger the emergency workflow manually.	M	P-08
FR-75	The system shall log the complete escalation sequence (trigger, prompt, response, notifications).	S	P-08

Scope note: The MVP demonstrates the escalation workflow using software-triggered and manually triggered events. It does not claim to detect incidents automatically (for example, loss of consciousness while driving), and does not integrate with real emergency services.

A.9 Consultation Support (Enhancements)

ID	Requirement	Priority	Problem link
FR-80	The system shall present a chronological health timeline.	C	P-01, P-09
FR-81	The system shall generate a doctor-visit summary of recent measurements, alerts and reported symptoms.	C	P-09
FR-82	The system shall support medication and appointment reminders.	C	P-05, P-09

A.10 Localization

ID	Requirement	Priority	Problem link
FR-90	The system shall provide the core user interface in Bangla and English.	M	P-06
FR-91	The system shall provide an accessibility mode for older adults (large text, high contrast, simplified navigation).	S	P-06

Appendix B: Requirements Traceability

This matrix links every problem (Section 3) to the objectives (Section 10) and requirements (Section 13, Appendix A) that address it. Parts 2 and 3 extend it with design components and test cases.

Problem	Objective(s)	Requirements
P-01 Fragmented information	O-01	FR-01 to FR-03, FR-10, FR-20 to FR-24, FR-30, FR-80
P-02 Limited personalization	O-02	FR-10, FR-11, FR-40, FR-41
P-03 Gradual changes unnoticed	O-02	FR-30 to FR-35
P-04 Family history disconnected	O-01, O-02	FR-12, FR-13
P-05 Habit consistency	O-01, O-02	FR-20, FR-23, FR-40 to FR-42, FR-82
P-06 Elderly monitoring gap	O-01, O-04	FR-04, FR-22, FR-60 to FR-66, FR-90, FR-91
P-07 Where to seek care	O-03	FR-50 to FR-55
P-08 Emergency communication gap	O-04	FR-66, FR-70 to FR-75
P-09 Consultation preparation	O-01	FR-80 to FR-82
Cross-cutting (security and privacy)	O-05	NFR-01 to NFR-05, FR-65
Cross-cutting (extensibility and scale)	O-05	NFR-10 to NFR-14

Appendix C: Glossary

Term	Definition
Alert level	A classification from Level 0 (Normal) to Level 4 (Emergency escalation) describing how far a reading or pattern deviates from expectation and what action follows
Escalation workflow	A configured sequence of software actions (verification prompt, contact notification, optional location sharing) started by a predefined event. It is not a clinical determination
Baseline	A user's own typical range of measurements, calculated from their history
Cold start	The period when there is too little data for a statistical or machine-learning method to be reliable
Guardrail layer	Software controls that filter and constrain the inputs and outputs of the AI assistant
Family-assisted entry	Data entry by an authorized caregiver on behalf of another user
Health and wellness companion	A non-diagnostic system that provides monitoring, guidance and support, and does not replace clinical care
MoSCoW	A prioritization method: Must, Should, Could, Won't (now)
MVP	Minimum Viable Product
Isolation Forest	An unsupervised machine-learning algorithm used to detect outliers
PDPO	Personal Data Protection Ordinance, 2025 (Bangladesh)
RBAC	Role-Based Access Control
SMART	Specific, Measurable, Achievable, Relevant, Time-bound

End of Part 1